Hilectro



Electric Forklift

Single and Double Drive Balanced Weight series

HAITIAN Drive Systems

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The Company reserves the final interpretation
to the upgrading of the parameters.

www.hilectro.com





HAITIAN Drive Systems

PROFILE

About Haitian Drive

Ningbo Haitian Drive Systems Co., Ltd. is one of the four pillar industries of Haitian Group. Relying on the advantages of Haitian Group platform, Haitian Drive has grown to be the leading manufacturer in the fields of servo control system (servo motor, servo drive, control system), automation (servo manipulator and peripheral automation, robot automation, magnetic clamping system, functional components), new energy (electric forklift) and hydraulic transmission (hydraulic motor, hydraulic pump and hydraulic components).

Haitian Drive: Innovation Drives Future

Haitian Drive put forward the “Innovation Drives Future” as the company philosophy in the very beginning based on the management of Haitian Group. We improve the product quality and try to perfect the technical service by continuous innovation with the aim of enhancing the overall competitiveness in the market.We firmly believe that products with superior quality can win more customers whose competitiveness will be enhanced in the meantime.

Hilectro

BRAND

The mature developing platform of Haitian Group

Relied on the mature developing platform of Haitian Group, Hilectro Power Technology has the self-developed controller, professional motor manufacturing team, large sheet metal and painting workshop . At the same time, the company carries out the ISO:9001 quality management systems strictly, pursuiting the quality improvement constantly .

Financial strength

The company is located in the Industry Zone of Beilun which is one of the biggest ports around the world. The area of the factories is covering 110 thous square meters. Equipped with multi production lines whose annual capacity is over 10 thous sets of forklifts, the company has the advanced testing center at the same time.

High quality R&D team

Gathering with the professional supervisors, the company has a high end R&D team. By combining the domestic and overseas experience, the team is improving the quality of the forklifts continuously, and trying to be ranked in the top suppliers in the industry.

State of the Art Electric Forklift Technology

Hilectro electrical forklift not only has the excellent performance which can compare with the combustion forklifts, but also has the green technology in low consumption. The user has zero burden to the environment.

We are trying to find the best solution in reducing the running expenses, improving the working efficiency, ensuring the safety and comforts, and supplying the customized service based on the working environment.

Satisfy the Green Logistics Demands Easily







Precise Design Reduce the performance Cost

Fully hydraulic minimum steering redius

The hydraulic power steering system of the Hilectro Electric Forlift is providing outstanding flexibility and comfort during the moving process. It can make the steering and the Z-shaped moving smoothly, easy and stable. With the small 300mm diameter steering wheel, the forklift has reached a very high sensitivity: undulation is minimized when moving straight which makes the driving stability even in the small space. Moreover, the forklift can automatically detect the drift of the steering wheel under the unsatisfied state and restore it to the original position.

> Easy, smooth and stable steering, Z-shaped driving
> Undulation reduction rapidly when driving straight
> Steering Sync function
> Minimum turning radius reaches 1750mm

Two AC drive motors, excellent drive performance

The two drives (each front wheel is driven by one AC motor) endues the Hilectro electric forklift great running performance which can compare with the combustion forklift. In fact, the system ensures the forklift excellent driving performance although on the floor which totally can not reach the requirements.

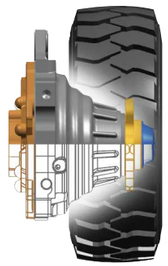
- Max speed **15km/h** (CPD15/ unloaded)
- Acceleration (the acceleration time from 0km/h to 5km/h) is **2.9S** (CPD15/ unloaded)
- Grade ability **22%** (CPD 15/ unloaded)

- Electronic Differential System in the left and right motors
- Good operating and firmly gripping ability
- Excellent acceleration **0km/h → 5Km/h** only need **3.8S**

Advanced wet brake system

Hermetically wet disc brake systems ensure the excellent overall durability effectively. Different with the conventional drum brake system, there is no need to replace the friction discs, does not require the regular maintenance, therefore reduce the maintenance costs.
The technology not only gives excellent control features in wet floor, but also improving the productivity.





AC CAN controller

By making the controller system lighter, simpler and more compact, Haitian Drive Systems pursues the best design continually. Besides the features as advanced controller (Adjust all the functionality as required) and security structures (for example, moving / lifting operator presence sensing system) features, the power saving function ensures the best performance of the AC motor even if in a high state of battery discharging.





The Superior Performance of High Efficiency



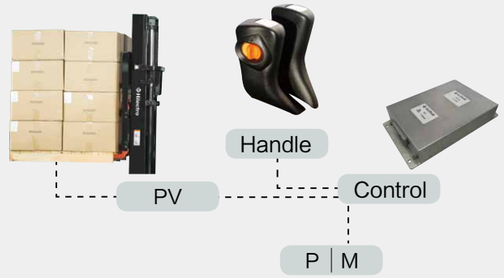
High performance battery

Customized battery proposal can provide 8 hour non-stop moving.

Controlled by the proportional hydraulic valve

In the process of moving, the accurate reliable computing systems control the fuel precisely, so as to providing a stable control and to minimize vibration. Therefore Hilectro electric forklift is the ideal equipment for the sensitive and fragile items.





Easy maintenance

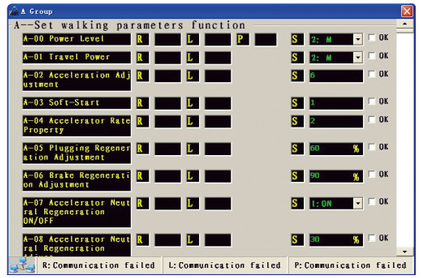
Hilectro electric forklift uses the auxiliary tipping battery cover which makes the battery check easier. There are side take and hanged out two ways. Moreover, the Grams of the Hilectrol original self-diagnostic system running on a standard PC greatly simplified the Problem-solving process, which minimized the downtime.

Automatic diagnosis system

The forklift automatic diagnosis system can detect the laden moving situation. the running situation of the drives and the laden running. Once a problem is found, the alarm will report it.

The outstanding steering stability

The design of high set rear axle and low-set battery ensures Hilectro electric forklift maintaining excellent stability even in high-speed steering. Meanwhile, the rubber underpass frame can absorb the tremor, which significantly improves the driving comfort.



Adjustable personal setting

The function of moving, lifting or tilting speed can be based on a variety of personal need which can be adjusted on the dashboard with simple steps easily. With the adjustable settings, the operation can be individually optimized, therefore providing more stable and more comfortable operation.

- > Driving speed setting·
- > lifting speed adjustment·
- > Reversal braking regeneration setting·
- > Tilting speed adjustment·
- > Accelerators median renewable setting



Warning display

	Battery level alarm lamp
(P)	Stop brake indicator light
(N)	Mid-level safety/driving interlock alarm light
	Abnormal alarm lamp
	The driving system abnormal alarm lamp
	Lifting system/lifting chain abnormal alarm lamp

Excellent Driving Safety And Comfort



The relentless pursuit of security and comfort

During the moving process, the most important thing is to reduce the total operating costs, and achieve the high productivity and high work efficiency, at the same time ensure the safety and comfort of the operator. With the features like hydraulic operation and driving control stick, function set, as well as the ergonomic layout, at the same time, the safety devices equipped with the operator presence sensing systems in order to ensure the safety of the operator at any time.

The ergonomic combined operating handle

Based on knowledge of ergonomics, the forklift has changed the traditional pre-mechanical operation, but optimized with the right set switch control which has reduced the labor intensity. In addition, the operator can control the forklift easily in the ideal posture, which ensures the comfort after the long time continuously working.





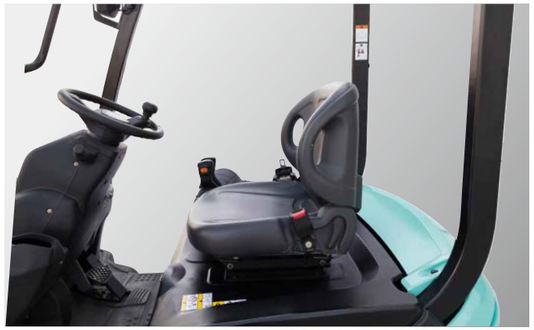
The best vision

For safety ,Hilectro electric forklift provides the best vision in all directions. In order to minimize the dead angle in the front side, the operator's seat is located in the left side. Moreover, the forklift also takes the advantages of the high visibility mast and the instrument panel on the top of the board to ensure a clear front-side vision, so that the operator can clearly see the fork tip.



OPS safety system

Driving/ rise OPS system is one of the safety functions which meet ISO3691. The safety system makes sure that only if the operator is on the seat the hydraulic or moving operation is allowed. When the operator leaves his seat without braking after 3 seconds, the alarm will be inspired and the operation stops. When the operator back to his seat, the alarm remains the stimulating state, the lock can be only released by return the switches to the safety position.



*Under the adverse conditions, for safety, the interlock function can stop driving, in order to avoid the wrong operation.

Easy get on and off

The step and the floor of Hilectro electric forklift is low and spacious. It can minimize on and off load of the operator. The carefully design around the steering and auxiliary handle makes the up and down of the forklift further conveniently, so the operator can move from one side to the other side easily.

HSS II seat for well body support

Hilectro original cushion and buffer technology effectively absorb vibration.And HSS, II seat specifically designed to support the entire body, It can minimize the fatigue even after long and continuous operation.

Adjust angle max to 70 degree
48mm suspension buffer design
170mm front/ back slide
Match the quality shrink seat belt

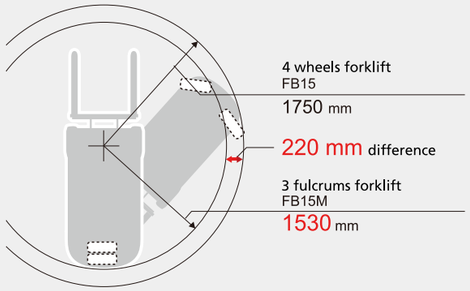


Three Fulcrums Forklift with Situ Cornering Ability

Three Fulcrums Forklift with Situ Cornering Ability

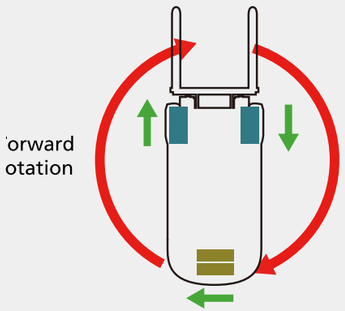
Three fulcrums forklift is the forklift which can cornering 90° in situ. What's more, when the back-wheels are cornering more than 75°,the inner wheel backwards function will only make the front inner wheel cornering in the opposite direction, which provides good driving force during the U-turing and situ turning.

Comparison in minimum turning radius





Full rotation in all directions







The Important Options



The Important Options

Digital load checker

Check the approximately load and display in unit of 10kg.



The laser height sensor

Use one laser beam points out the height of the forks, which makes the forks insert the platen easier and more stable.

Resin top cover

With the protection from the resin top cover, the forklift can even drive in the rain.

The left/right rearview mirror

Besides the central rearview mirror, the hilectro forklift equipped with the left and right rearview mirrors to make sure the safety working.

Personalized Customization

Forklift Options

Cold storage option (-35°C)

The anti-corrosion coating and durability makes the forklifts be reliable equipment even in the harsh production condition.

Fisheries option

The anti-corrosion coating and durability makes the forklifts performance well in the salty and water condition.

Anti-dust (electrical seal) option

All main electrical parts are sealed compactly to isolate the dust. This option conducive the forklift working in the dusty condition.

The Door Frame

2 grades door frame with free vision

(Max forks lifting height: 2700-5000mm) Featured with the wide front vision between the inner door frame.

Door frame with free vision

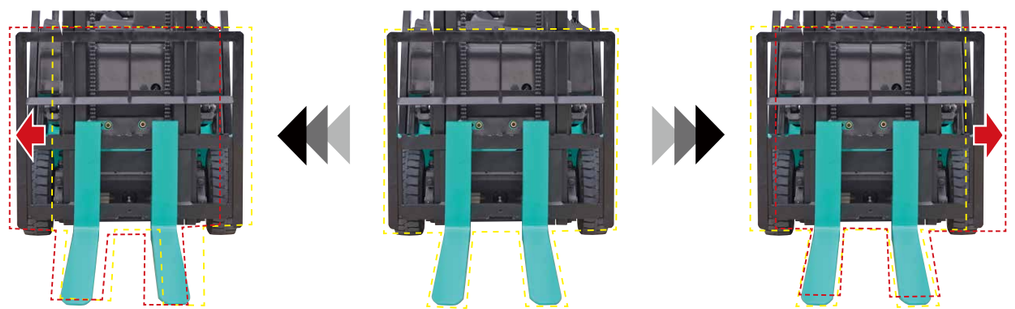
(Max forks lifting height: 3000-4000mm) The frame with big lifting ability, especially suitable for the frame-limit application.

3 grades door frame with free vision

(Max forks lifting height: 3700-6000mm) With higher max forks lifting height, especially suitable for operation which lifting the load to the high height.

Working Equipment

- Integral sway device
- Forks locator
- Scroll staples
- Hinged forks
- Load stabilizer
- Forks staples
- Strapping staples



Standard Specifications

Balanced Weight Electric Forklift
1.5 ton --3.5 ton









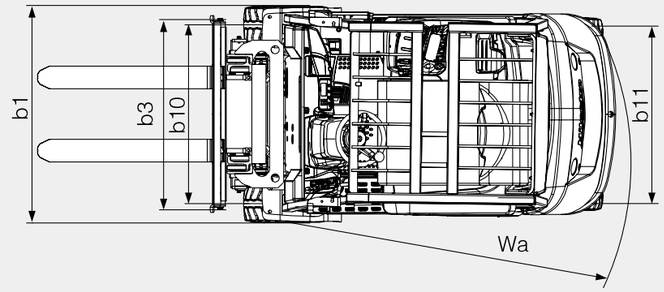
						CPD15M	CPD18M	CPD15	CPD20	CPD25	CPD30	CPD35
parameter	1.2	Model	Manufacturers grades			Electric	Electric	Electric	Electric	Electric	Electric	Electric
	1.3	Power modes	Ele ctric, Diesel, Petrol			Sit	Sit	Sit	Sit	Sit	Sit	Sit
	1.4	Operate mode				1500	1800	1500	2000	2500	3000	3500
	1.5	Rated capacity	Q	Rated capacity	kg	500	500	500	500	500	500	500
	1.6	Load center	c	Rated load center	mm	1350	1570	1350	1800	2250	2700	3150
	1.6.1	Optional capabilities	Q2	capability at 600mm load center	kg	435	435	435	435	485	485	490
	1.8	Load distance	x	Front axle center to the fork surface	mm	1340	1485	1410	1470	1720	1720	1720
	1.9	Wheelbase	y		mm	2980	3730	2930	3620	4750	5000	5350
Weight	2.1	Use weight	Including the smallest capacity battery, see 6.5 line		kg	4000	5080	3980	5020	6450	7168	7920
	2.2	Axle Loading	Load	Front	kg	480	650	450	600	800	832	930
	2.2.1			Back	kg	1460	1790	1490	1720	2150	2350	2427
	2.3		Unload	Front	kg	1520	1940	1440	1900	2600	2650	2923
	2.3.1			Back	kg	Pneumatic tires	Pneumatic tires	Pneumatic tires	Solid tires	Solid tires	Solid tires	Solid tires
Tire	3.1	Tyre Type	Pneumatic tires, solid tires			18×7-8	18×7-8	18×7-8	200/50-10	23×10-12	23×10-12	23×10-12
	3.2	Tyre Size	Front			15×41/2-8	15×41/2-8	16×6-8	16×6-8	18×7-8	200/50-10	200/50-10
	3.3		Back			2/2	2/2	2/2	2/2	2/2	2/2	2/2
	3.5	Number of Wheel	Front/Back			940	950	940	950	1100	1100	1100
	3.6	Tread,Front	b10		mm	170	170	865	865	980	980	980
	3.7	Tread,Rear	b11		mm	6/10	6/10	6/10	6/10	6/10	6/10	6/10
Size	4.1	Tilting Angle	a/b	Forward/backward	degree	2050	2050	2050	2050	2142	2142	2142
	4.2	Mast Height,Lowered	h1	2 grade frame	mm	140	140	140	140	120	120	120
	4.3	Std.Free Lift	h2	2 grade standard door frame, from ground	mm	3000	3000	3000	3000	3000	3000	3000
	4.4	Std.Lift Height	h3	2 grade standard door frame, from ground	mm	3955	3955	3955	3955	4279	4279	4279
	4.5	Mast Height,Extended	h4	2 grade standard door frame,	mm	2070	2070	2070	2070	2215	2215	2215
	4.7	Height, Overhead Guard	h6		mm	2885	2885	3010	3250	3550	3550	3550
	4.19	Length,with Std.Forks	l1		mm	1935	1935	2060	2180	2480	2480	2480
	4.20	Length,to Fork face	l2		mm	1110	1110	1110	1160	1360	1360	1360
	4.21	Width at Type	b1	single	mm	35×100×950	40×120×1070	35×100×950	40×120×1070	45×125×1070	45×125×1070	50×125×1070
	4.22	Forks	s/e/l		mm	Class2，Type A	Class2，Type A	Class2，Type A	Class2，Type A	Class3，Type A	Class3，Type A	Class3，Type A
	4.23	Forks Carriage Class	ISO 2328,Type A/B/no			970	970	970	970	1100	1100	1100
	4.24	Width carriage	b3		mm	124	124	124	124	142	142	142
	4.31	Ground Clearance	m1	under the door frame	mm	125	125	125	125	130	130	130
	4.32		m2	Center of wheelbase	mm	2935	3095	3155	3305	3645	3645	3645
	4.33	Right Angle Stacking Aisle	Ast	with L1000×W1200 bracket	mm	3135	3295	3355	3505	3845	3845	3845
	4.34		Ast	with L1200×W800 bracket	mm	1550	1600	1750	1850	2200	2230	2250
	4.35	Turning Radius	Wa		mm	loaded/ unloaded	14/15	14/15	14/15	14/15	15/16	14/15	14/15
Perfor-mance	5.1	Travel Speed	loaded/ unloaded		km/h	340/560	265/450	340/560	265/450	370/400	350/400	260/340
	5.2	Lifting Speed	loaded/ unloaded		mm/s	450/380	430/340	450/380	430/340	390/320	390/320	340/300
	5.3	Lowering Speed	loaded/ unloaded		mm/s	Load, 3min, rated	10000	10000	10000	10000	19000	19000	19000
	5.6	Max.Drawbar Pull	Load, 3min, rated		N	no-Load, 3min, rated	7500	7500	7500	7500	15500	15500	15500
	5.7		no-Load, 3min, rated		N	loaded,3min,rated/1.5/hour	19	17	19	17	20	19	17
	5.8	Max.Gradeability	loaded,3min,rated/1.5/hour		%	operate/control pedal/hydraulic	pedal/hydraulic	pedal/hydraulic	pedal/hydraulic	pedal/hydraulic	pedal/hydraulic	pedal/hydraulic	pedal/hydraulic
	5.10	Service Brake	operate/control pedal/hydraulic			operate/control manual/mechanical	manual/mechanical	manual/mechanical	manual/mechanical	manual/mechanical	manual/mechanical	manual/mechanical	manual/mechanical
	5.10.1	Parking Brake	operate/control manual/mechanical			pattern	FHPS	FHPS	FHPS	FHPS	FHPS	FHPS	FHPS
	5.10.2	Steering	pattern			60min, rating	4.5×2	4.5×2	4.5×2	4.5×2	10×2	10×2	10×2
Drive	6.1	AC Drive Motor (AC)	60min, rating		kw	5min, rating	9	9	9	9	20	20	20
	6.2	AC Pump Motor (AC)	5min, rating		kw	48	48	48	48	48	80	80	80
	6.4	Battery Voltage			V	Ah/5hr	460	460	460	460	560	560	560
	6.4.1	Rated Battery Capacity			Ah/5hr	Ah/5hr	390	390	390	390	460	460	460
	6.4.2	Battery Capacity , min			Ah/5hr	Ah/5hr	560	560	560	560	630	630	630
	6.4.3	Battery Capacity , max			kg	750	750	750	750	750	1300	1300	1300
	6.5	Battery Weight				Transistors	Transistors	Transistors	Transistors	Transistors	Transistors	Transistors	Transistors
others	8.1	Drive Motor Control			bar	176	176	176	176	176	176	176	176
	8.2	Relief Pressure Attachment			L	23	23	23	23	50	50	50	50
	8.2.1	Tank Capacity				Integrated	Integrated	Integrated	Pluggable type	Pluggable type	Integrated	Pluggable type	Pluggable type
	8.3	Traction Pin Installation Mode	Integrated ／ Pluggable type										

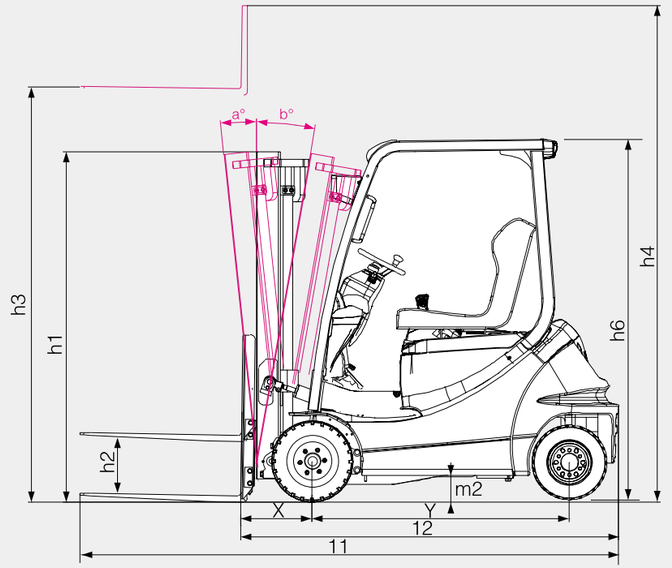
Main Parts

（●：Standard ○：Options △：Delivery as the customer's requirements ： N/A S：Adjust by the maintenance technician）

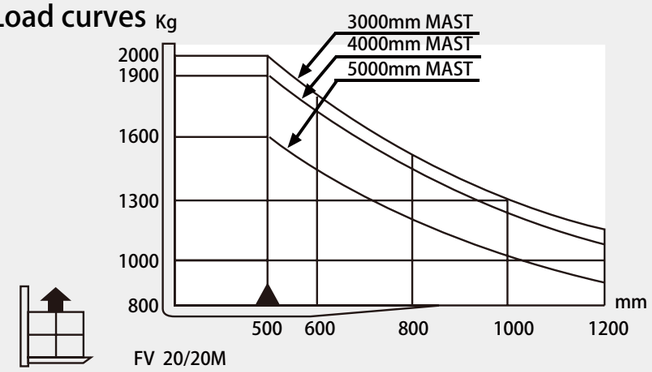
			Standard forklift 1.5-3.5t	Standard forklift 2.5-3.5t	Standard forklift 1.5-1.8t
ICS			●	●	●
Motor		Driving motor	AC	AC	AC
		Hydraulic and cornering motor	AC	AC	AC
Double motor drive			●	●	●
Hermetically wet disc brake system			●	●	●
Proportional hydraulic valve			●	●	●
Combined Operation Rod (integrated advance, backward switch)			●	●	●
Turning mode			FHPS	FHPS	FHPS
Setting	Driving mold adjustment	Driving speed	●	●	●
		Reverse brake regeneration	● (S)	● (S)	● (S)
		Brake regeneration	● (S)	● (S)	● (S)
		Soft start	● (S)	● (S)	● (S)
		Accelerators	● (S)	● (S)	● (S)
		Accelerator mid-position regeneration	● (S)	● (S)	● (S)
		Regenerating Tilt Angle	● (S)	● (S)	● (S)
		Speed Control	●	●	●
	Hydraulic Control Mode Adjustment	Lifting speed adjustment	●	●	●
		Tilting speed adjustment	● (S)	● (S)	● (S)
		Working equip speed	● (S)	● (S)	● (S)
Regeneration		Reverse brake regeneration	●	●	●
		Brake regeneration	●	●	●
		Accelerator mid-position regeneration	●	●	●
IMS		Speed table	●	●	●
		Forward/backward indicator	●	●	●
		Speed limitation indicator	●	●	●
		Calendar/hour display	●	●	●
		Driving power display	●	●	●
		Battery discharge indicator	●	●	●
		Forks level indicator	○	○	○
		Load checker (Unit 10kg)	○	○	○
		Brake indicator	●	●	●
		Mid-position safety indicator	●	●	●
		Driving OPS alarming light	●	●	●
		Lifting OPS alarming light	●	●	●
		Error indicator	●	●	●
		Electrolyte level indicator	●	●	●
Control equipment related parts		Telescopic tilt steering column	●	●	●
		Automatic back turn signal lever	●	●	●
		Soft landing device	●	●	●
Safety support device		The operator presence sensing system (Lifting/driving Interlock device)	●	●	●
		Urgent switch	●	●	●
		Mid-position safety	●	●	●
		Auto power off	●	●	●
		Anti-rollback	●	●	●
		Driving speed control	●	●	●
		Key turn off lift lock	●	●	●
		Parking brake with release button	●	●	●
		Reversing buzzer	●	●	●
		Load checker	○	○	○
		Forward/backward alarming	○	○	○
		Rearview mirror (Left and right)	○	○	○
		Wide-angle central lens	●	●	●
		Back working lights	○	○	○
		flash lights (yellow/red/blue)	○	○	○
		Fire extinguisher	○	○	○
Support device used in the hydraulic control		Automatic fork leveling device	○	○	○
		Laser lifting height sensor	○	○	○
		Auto lifting stop function	△	△	△
		Soft carrying (hydraulic accumulators)	○	○	○
External parts		Back platen mats	●	●	●
		HSS II seat (with safety belt)	●	●	●
		Steel wheel house	△	△	△
		Windshield wipers (with window washer)	○	○	○
		Protection resin top cover	○	○	○
		Seat heater	○	○	○
The other		Pen barrel	●	●	●
		File folder	●	●	●
About battery		Batter side in and out	●	●	●
		Easy opening side door	○	○	○
		Battery with easy water providing device	○	○	○

■ Size





■ Load curves Kg



Gantry Specifications

Balanced Weight Electric Forklift
1.5 ton --3.5 ton





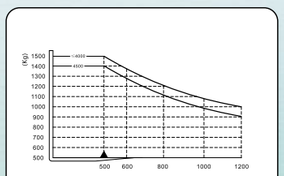




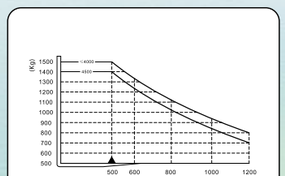
CPD15-20 Gantry Specifications

Gantry type	Product description	Maximum fork height	Gantry height			Free lifting height		The lowest ground clearance of the gantry	Gantry dip angle		Lifting capacity	
			Landing height of the gantry	Extending height of the gantry		Without load-backrest	With load-backrest				500mm load center distance	
				Without load-backrest	With load-backrest				Nose down	Hypsokinesis	Four supporting points	Three fulcrums forklift
		mm	mm	mm	mm	mm	mm	mm	Deg	Deg	Kg	Kg
CPD15/15M gantry											1.5T	1.5T
	Level II standard gantry	3000	2050	3600	3950	135	135	124	6	10	1500	1500
		3300	2200	3900	4250	135	135	124	6	10	1500	1500
		3500	2300	4100	4450	135	135	124	6	10	1500	1500
		3700	2400	4300	4650	135	135	124	6	10	1500	1500
		4000	2600	4600	4950	135	135	124	6	10	1500	1500
		4500	2850	5100	5450	135	135	124	6	6	1400	1400
	Level II free gantry	3000	2025	3600	3950	1450	1100	124	6	10	1500	1500
		3300	2175	3900	4250	1600	1250	124	6	10	1400	1350
		3500	2275	4100	4450	1700	1350	124	6	10	1350	1250
	Level III free gantry	4500	2050	5100	5450	1475	1125	124	6	6	1300	1350
		5000	2250	5600	5950	1675	1325	124	6	6	1200	1250
		5500	2500	6100	6450	1925	1575	124	3	3	1000	1100
		6000	2690	6600	6950	2115	1765	124	3	3	800	100
CPD20/18M gantry											2T	1.8T
	Level II standard gantry	3000	2050	3600	3950	140	140	124	6	10	2000	1800
		3300	2200	3900	4250	140	140	124	6	10	2000	1800
		3500	2300	4100	4450	140	140	124	6	10	2000	1800
		3700	2400	4300	4650	140	140	124	6	10	2000	1800
		4000	2600	4600	4950	140	140	124	6	10	2000	1800
		4500	2850	5100	5450	140	140	124	6	6	1800	1750
	Level II free gantry	3000	2025	3600	3950	1455	1105	124	6	10	2000	1800
		3300	2175	3900	4250	1605	1255	124	6	10	1900	1650
		3500	2275	4100	4450	1705	1355	124	6	10	1850	1550
	Level III free gantry	4500	2050	5100	5450	1480	1130	124	6	6	1600	1550
		5000	2250	5600	5950	1680	1330	124	6	6	1300	1250
CPD20/18M gantry (Import box iron)	Level III free gantry	4500	2024	5090	5450	1464	1104	124	6	6	1600	1550
		5000	2224	5590	5950	1664	1304	124	6	6	1300	1250
		5500	2522	6090	6450	1962	1602	124	3	3	1100	1050
		6000	2714	6590	6950	2154	1794	124	3	3	1000	950

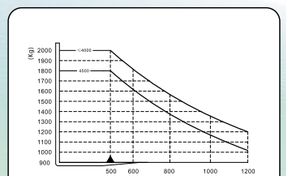
Load Curve Graph



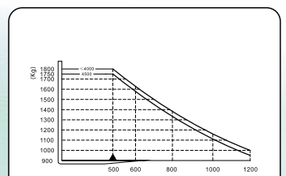
Four supporting points CPD15 load center distance (mm)
Warning: The rated load capacity of the accessories with sideway needs to decrease 150kg!



Four supporting points CPD15M load center distance (mm)
Warning: The rated load capacity of the accessories with sideway needs to decrease 150kg!



Four supporting points CPD20 load center distance (mm)
Warning: The rated load capacity of the accessories with sideway needs to decrease 150kg!

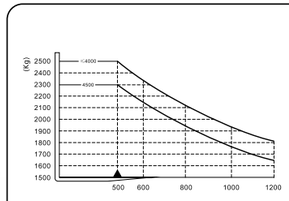


Four supporting points CPD18M load center distance (mm)
Warning: The rated load capacity of the accessories with sideway needs to decrease 150kg!

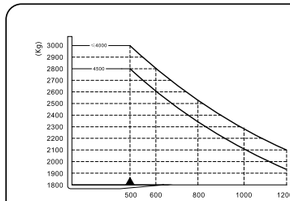
CPD25-35 Gantry Specifications

Gantry type	Product description	Maximum fork height	Gantry height			Free lifting height		The lowest ground clearance of the gantry	Gantry dip angle		Lifting capacity		
			Landing height of the gantry	Extending height of the gantry		Without load-backrest	With load-backrest		Nose down	Hypsokinesis	500mm load center distance		
				Without load-backrest	With load-backrest						Four supporting points		
		mm	mm	mm	mm	mm	mm	mm	Deg	Deg	Kg	Kg	Kg
CPD25/30/35 gantry											2.5T	3T	3.5T
	Level II standard gantry	3000	2142	3681	4279	140	140	142	6	10	2500	3000	3500
		3300	2292	3981	4579	140	140	142	6	10	2500	3000	3500
		3500	2392	4181	4779	140	140	142	6	10	2500	3000	3500
		3700	2492	4381	4979	140	140	142	6	10	2500	3000	3500
		4000	2692	4681	5279	140	140	142	6	10	2500	3000	3500
		4500	2942	5181	5779	140	140	142	6	6	2300	2800	3000
	Level II free gantry	3000	2112	3691	4281	1455	865	142	6	10	2500	3000	3500
		3300	2262	3991	4591	1605	1015	142	6	10	2400	2900	3400
		3500	2362	4191	4781	1705	1115	142	6	10	2300	2800	3300
	Level III free gantry	4500	2112	5197	5787	1455	865	142	6	6	2000	2600	2800
		5000	2312	5697	6287	1655	1065	142	6	6	1600	2200	2500
		5500	2474	6197	6787	1818	1228	142	3	3	1300	1800	2000
		6000	2762	6697	7287	2105	1515	142	3	3	1150	1200	1500

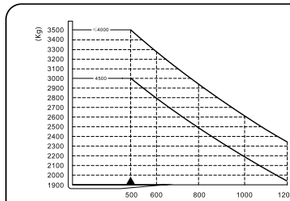
Load Curve Graph



Four supporting points CPD25 load center distance (mm)
Warning: The rated load capacity of the accessories with sideway needs to decrease 200kg!



Four supporting points CPD30 load center distance (mm)
Warning: The rated load capacity of the accessories with sideway needs to decrease 200kg!



Four supporting points CPD35 load center distance (mm)
Warning: The rated load capacity of the accessories with sideway needs to decrease 250kg!



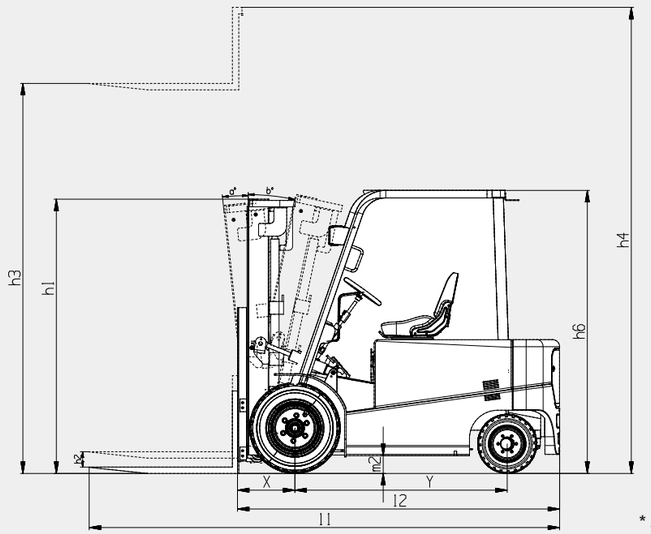












* 以 CPD30KJ 为例

parameter	1.2	Model	Manufacturers grades				CPD15K	CPD20K	CPD25K	CPD30K	CPD35K	CPD40K	CPD45K	CPD50K
	1.3	Power modes	Ele ctric, Diesel, Petrol				Electric	Electric	Electric	Electric	Electric	Electric	Electric	Electric
	1.4	Operate mode					Sit	Sit	Sit	Sit	Sit	Sit	Sit	Sit
	1.5	Rated capacity	Q	Rated capacity		kg	1500	2000	2500	3000	3500	4000	4500	5000
	1.6	Load center	c	Rated load center		mm	500	500	500	500	500	500	500	500
	1.6.1	Optional capabilities	Q2	capability at 600mm load center		kg	1350	1800	2250	2700	2700	3850	4100	4800
	1.8	Load distance	x	Front axle center to the fork surface		mm	400	430	430	475	490	560	560	560
	1.9	Wheelbase	y			mm	1350	1448	1448	1652	1652	2030	2030	2030
Weight	2.1	Use weight	Including the smallest capacity battery, see 6.5 line			kg	2950	3812	4120	5170(Pneumatic tires)	5520	6350	6750	7080
	2.2	Axle Loading	Load	Front		kg	4000	5020	5958	7170	7960	9100	10080	10800
	2.2.1			Back		kg	450	600	662	1150	1060	1250	1270	1280
	2.3		Unload	Front		kg	1510	1721	1854	2460	2560	3120	3180	3280
	2.3.1			Back		kg	1440	2091	2266	2800	2960	3230	3530	3800
Tire	3.1	Tyre Type	Pneumatic tires, solid tires				Solid tires	Pneumatic tires	Pneumatic tires	Pneumatic tires /Pneumatic tires	Solid tires/Solid tires	Solid tires	Solid tires	Solid tires
	3.2	Tyre Size	Front				6.00-9	7.00-12	7.00-12	28*9-15	28X9-15	28*12.5-15	28*12.5-15	28*12.5-15
	3.3		Back				16X6-8	18×7-8	18×7-8	18*7-8	200x50-10	21*8-9	21*8-9	21*8-9
	3.5	Number of Wheel	Front/Back				2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2
	3.6	Tread,Front	b10			mm	900	992	992	1008	1008	1150	1150	1150
	3.7	Tread,Rear	b11			mm	920	960	960	978	978	1080	1080	1080
Size	4.1	Tilting Angle	a/b	Forward/backward		degree	6/12	6/12	6/12	6/12	6/12	6/12	6/12	6/12
	4.2	Mast Height,Lowered	h1	2 grade frame		mm	2065	2055	2055	2145	2145	2195	2195	2195
	4.3	Std.Free Lift	h2	2 grade standard door frame, from ground		mm	56	140	140	160	160	120	120	120
	4.4	Std.Lift Height	h3	2 grade standard door frame, from ground		mm	3000	3000	3000	3082	3082	3060	3060	3060
	4.5	Mast Height,Extended	h4	2 grade standard door frame,		mm	3955	4032	4032	4285	4285	3130	3130	3130
	4.7	Height, Overhead Guard	h6			mm	2084	2154	2154	2215	2215	2150	2150	2150
	4.19	Length,with Std.Forks	l1			mm	2931	3400	3400	3679	3679	4130	4130	4130
	4.20	Length,to Fork face	l2			mm	2011	2330	2330	2560	2560	3020	3020	3050
	4.21	Width at Type	b1	single		mm	1120	1193	1193	1200	1200	1445	1445	1445
	4.22	Forks	s/e/l			mm	35×100×950	40×120×1070	40×120×1070	50×125×1070	50×125×1070	50*150*1070	50*150*1070	50*150*1070
	4.23	Forks Carriage Class	ISO 2328,Type A/B/no				Class2，Type A	Class2，Type A	Class2，Type A	Class3，Type A	Class3，Type A	Class3，Type A	Class3，Type A	Class3，Type A
	4.24	Width carriage	b3			mm	970	1038	1038	1100	1100	1380	1380	1380
	4.31	Ground Clearance	m1	under the door frame		mm	115	112	112	120	120	106	106	106
	4.32		m2	Center of wheelbase		mm	125	106	106	125	125	140	140	140
	4.33	Right Angle Stacking Aisle	Ast	with L1000×W1200 bracket		mm	3385	3770	3770	4030	4030	4450	4450	4450
	4.34		Ast	with L1200×W800 bracket		mm	3585	3900	3900	4060	4060	4580	4580	4580
	4.35	Turning Radius	Wa			mm	1900	2250	2250	2350	2350	2680	2680	2700
Perfor-mance	5.1	Travel Speed	loaded/ unloaded			km/h	13/15	14/15	14/15	14/15	14/15	13/16	13/16	13/16
	5.2	Lifting Speed	loaded/ unloaded			mm/s	340/560	280/350	270/360	310/405	270/400	320/420	320/420	320/420
	5.3	Lowering Speed	loaded/ unloaded			mm/s	450/380	420/500	420/500	400/480	400/480	420/420	420/420	420/420
	5.6	Max.Drawbar Pull	Load, 3min, rated			N	11000	10700	10700	13000	13000	22000	22500	23000
	5.7		no-Load, 3min, rated			N	8000	10000	10000	14000	14000	20500	20500	20500
	5.8	Max.Gradeability	loaded,3min,rated/1.5/hour			%	16	15	15	15	12	15	15	15
	5.10	Service Brake	operate/control pedal/hydraulic				pedal/hydraulic	pedal/hydraulic	pedal/hydraulic	pedal/hydraulic	pedal/hydraulic	pedal/hydraulic	pedal/hydraulic	pedal/hydraulic
	5.10.1	Parking Brake	operate/control manual/mechanical				manual/mechanical	manual/mechanical	manual/mechanical	manual/mechanical	manual/mechanical	manual/mechanical	manual/mechanical	manual/mechanical
	5.10.2	Steering	pattern				Hydraulic steering	Hydraulic steering	Hydraulic steering	Hydraulic steering	Hydraulic steering	Hydraulic steering	Hydraulic steering	Hydraulic steering
Drive	6.1	AC Drive Motor (AC)	60min, rating			kw	8	8	8	15	15	18	18	18
	6.2	AC Pump Motor (AC)	5min, rating			kw	8.6	8.6	8.6	12	12	25.4	25.4	25.4
	6.4	Battery Voltage				V	48	48	48	80	80	80	80	80
	6.4.1	Rated Battery Capacity				Ah/5hr	390	630	630	600	600	730	730	730
	6.4.2	Battery Capacity , min				Ah/5hr	385	480	480	460	460	730	730	730
	6.4.3	Battery Capacity , max				Ah/5hr	455	640	640	640	640	1000	1000	1000
	6.5	Battery Weight				kg	600	950	950	1400	1400	2000	2000	2000
others	8.1	Drive Motor Control					MOSFET/AC	MOSFET/AC	MOSFET/AC	MOSFET/AC	MOSFET/AC	MOSFET/AC	MOSFET/AC	MOSFET/AC
	8.2	Relief Pressure Attachment				bar	176	176	176	176	176	220	220	220
	8.2.1	Tank Capacity				L	23	25	25	40	40	50	50	50
	8.3	Traction Pin Installation Mode	Integrated ／ Pluggable type				Pluggable type	Pluggable type	Pluggable type	Integrated	Integrated	Pluggable type	Pluggable type	Pluggable type

CPD15/20/25/30K

Appearance

The appearance modeling adopts the design method of the modern car, i.e. the international popular arc, the streamline, the surface modeling, the beautiful surface, conform to the modern esthetic effect and ergonomics, and the expression of the Haitian's characters.

Comfort

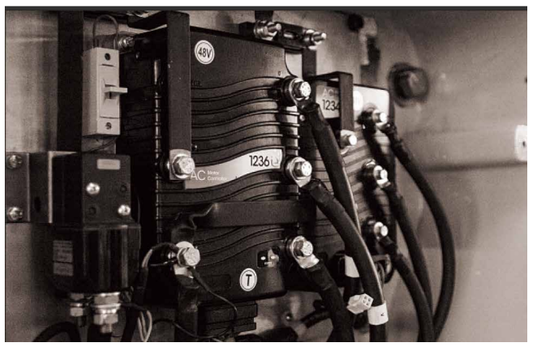
The inclining adjustable type steering wheel shall be designed in accordance with the ergonomics, the seat can be adjusted within a range of 150mm from front to back, and the operator can choose the best driving position to efficiently alleviate the driver's fatigue strength.
The gantry, adopting the optimized design, can make the lifting and declining more stable. The enlarged file open size of the carriage will extend the view field.
The declining and damping is of intelligent perception. When the fork is 100-60mm above the ground, the declining speed will be automatically slowed down, to avoid the goods and tray from bumping to the ground and protect the valuable goods and ground.
The new integrated combination instrument is complete in functions and direct in reading.
The advantage of silence, no non-pollution and energy saving coincides with the environmental requirements.



Advancement

The high-frequency CURTIS integrates with controller to guarantee the stable and accurate control of the driving and lifting, the excellent speed adjustment capacity, the functions of plug braking and anti-sliding in the slope, and more efficient, safer and more comfortable operation.
Adopt the big screen LED electric quantity display meter, chronometer and fault self-inspection meter of the international top brand that can undertake an accurate display under the severe environment
The international advanced CURTIS controller achieves the computing control in the walking and lifting of the fork lift truck.
Adopt the CAN bus communication technology, check the information of motor speed and the temperature, etc. in real time, and achieve the closed-loop control.
Adopt the high-power AC walking motor and the oil pump motor, high in the system efficiency, complete in the protection function, autonomous sensors and improved reliability and serving life.
The integral-type wet-type braking bridge, standard configures the standard wet-type braking, the braking system enjoy a lifelong free maintenance.
Adopt the new-type tooth graduation technique and low-noise gear oil pump.





Serviceability

The opening width of the cover of the storage battery is big (with gas spring assistance), making the maintenance of the storage battery very convenient. The closed electronic control system design can prevent the electronic control system from being eroded by the rain and the dust.
IP54 Level walking motor is dustproof, waterproof and high in reliability.
Pump motor, pump, oil tank and electronic control system of the single driving fork lift trucks of 2 tons and 2.5 tons all adopt the new-type left, right and trailer arrangement structure (three-gate type, convenient in the maintenance, excellent in waterproof, dust proof and noise reduction.)
The single driving fork lift truck of 3 tons adopts the domestic advanced hydraulic system configuration structure, the creative integrated oil box, and a tight structure, at the same time the convenience of the maintenance of the hydraulic system and the electronic system is guaranteed.

Reliability, safety

The main electrical components such as motor controller, contactor, power plug, emergency outage switch, instrument panel and accelerator, etc. all adopt the international famous brand products.
The emergency power off switch as the standard configuration and comply with the European safety standards.
AC walking motor and oil pump motor as the standard configuration, the oil pump motor may choose the DC pump motor. The electronic load protection device.

The advantages of the AC control system

The AC motor is maintenance-free, without wearing parts like carbon brush and brush carrier, etc. that need to be regularly maintained, low maintenance fee.
The AC power system is high in efficiency, complete in protection function, high in reliability and long in serving life.
The anti-sliding function that can't be realized by the series excitation electronic control can be achieved.
The multi-contactor of the series excitation electronic control can be left out (such as the turning, the bypass, and the regeneration contactor).
The fluctuation of the battery and the motor working current is reduced, while the reliability and the serving life are highly improved.



Introduction of the single driving product of 1 ton

Creative side-opening type door structure design

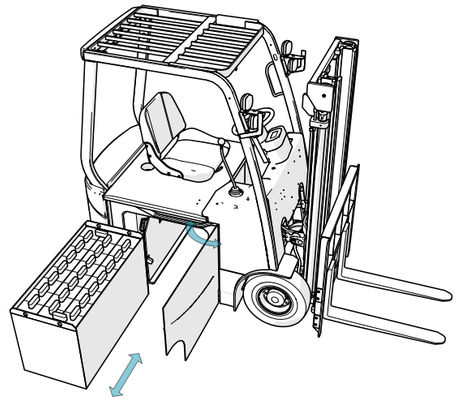
The side-opening type door design matches up with the manual hydraulic power unit and the simple tooling. A person can finish the storage battery exchange in 3 minutes.
No need for any assistance from the lifting appliance like crane and fork lift truck, it's safe, convenient and effort-less in the operation.
The storage battery locates at the bottom part of the vehicle center, the product gravity is low, and it's stable under the turning and slop situations.

Super wide view field for the gantry design

The external and internal gantry adopt the H shape steel. Compared to the traditional gantry, the driver is provided a wider view field and improved driving safety.
Especially for the free and accessory gantry, the view is improved obviously, and the operation is safer and more efficient.

The integrated operation handle

The lifting, nosing down and hypsokinesis functions of the gantry adopt the single-bar integrated manual operation handler, it is tidy, beautiful and efficient.





CPD40/45/50K

Appearance

Adopt the international popular arc, streamline, surface modeling, the appearance is beautiful and with a stronger streamline.

Comfort

Steering wheel: The inclining adjustable type small steering wheel is designed in accordance with the ergonomics design, so the driver enjoys a good driving experience. Under the steering wheel are the light switch, the instrument display panel, the vehicle power switch, with a tight and reasonable structure, and through the CAN bus connect to the electronic control, the reliability is excellent. Chair: ① The seat can be adjusted within a range of 150mm from front to back, the adjustment range of the backrest is ±30°, the weight adjustment range is 45-160kg, cooperates with the integrated seat arm rest, the operator can choose the best driving position.

② The right seat arm rest can undertake an up and down and a back and forth adjustment, with jogging type operation handle switch (controlled by fingers), the emergency power-off switch, the forth and back switches to make it convenient for the driver to control the whole vehicle comfortably. ③ The suspension cushioning design, the cushioning and cushioning technology can effectively absorb the vibration. What's more, the seats are specially designed to support the whole body, even a continuous operation process for a long term can minimize the fatigue. ④ Equip with two-point-type high quality ETR to maximize the protection of the driver. The put-in type waterproof file bag is equipped on the back of the seat to collect and store the files. ⑤ The OPS safety system will allow the hydraulic pressure or the driving operation only when the driver sits down. The fork lift truck function will be cut off if the driver leaves the seat. It's of high safety.

Operation handle: The operation handle adopts the jogging type switch which locates above the right hand rail of the seat and is convenient for the operator control comfortably (two handle sizes for choosing).



Gantry: ① The gantry, adopting the optimized design, can make the lifting and declining more stable. ② The enlarged file open size of the carriage will extend the view field. ③ The declining and damping is of intelligent perception. When the fork is 100-60mm above the ground, the declining speed will be automatically slowed down, to avoid the goods and tray from bumping to the ground and protect the valuable goods and ground.

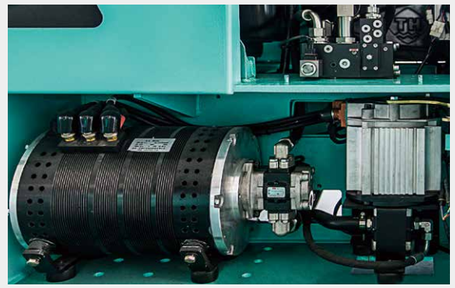


Instrument: The new-typed integrated-type combination instrument is complete in function, direct in reading, fast in parameter setting, convenient and fast. The location of the driver's instrument panel is moved upon the instrument stand, easy to view by lowering the head and convenient to operate the button choosing.

The Advanced Nature



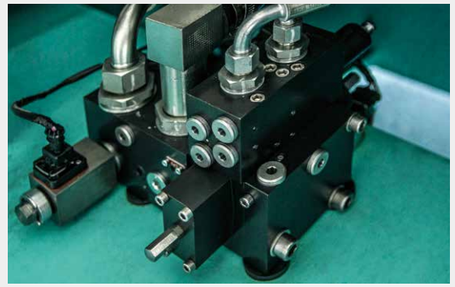
Brake: The column-assist type brake scheme, the enlarged brake pedal and the hydraulic power braking function, can effectively relieve the driver's fatigue strength, and make the brake more efficiently.



Adopt the dual pump hydraulic system, independent powers for lifting and braking, equipped with AC motor, make the hydraulic system stable and perfect. Equipped with the new-type tooth graduation technique and low-noise gear oil pump.

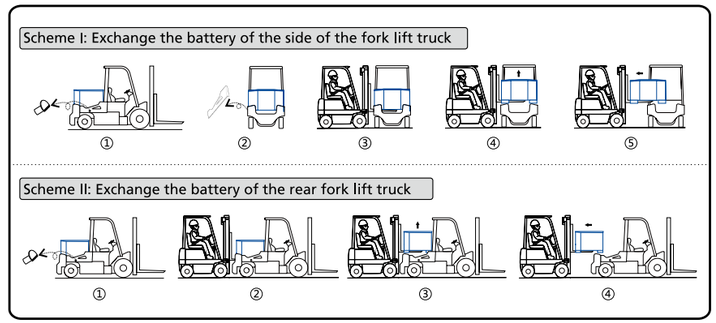


The high-frequency ZAPI integrates with controller to guarantee the stable and accurate control of the driving and lifting, the excellent speed adjustment capacity, the functions of plug braking and anti-sliding in the slope, and more efficient, safer and more comfortable operation. The international advanced ZAPI controller achieves the computer control in driving, lifting and turning of the fork lift truck.
Adopt the CAN bus communication technology, check the information of motor speed and the temperature, etc. in real time, and achieve the closed-loop control.



Adopting the international advanced Germany imported electro-hydraulic proportional valve, the computing systems can accurately control the fuel capacity during the operation process of the fork lift truck, so as to provide a stable control and minimize the vibration. Therefore, it's the ideal equipment to process the sensitive fragile. Can achieve the jogging-moving precise control of the gantry.

Reliability, safety, convenience



> The storage battery can be exchanged by the method of the military fork, the rear fork and the hoisting, and it's convenient for maintenance.
> The creative door-type structure, with a door in each side under the vehicle frame of the fork lift truck, to make the maintenance of the motor, the oil pump and the electronic control system more convenient, waterproof and dustproof, and low in noise.

The battery adopts postposition design, the cover plate is attached on the top of the battery, at the same time of guarantee the waterproof of the battery, make the battery maintenance quite convenient.

> The main electrical components such as motor controller, contactor, power plug, emergency outage switch and instrument panel, etc. all adopt the overseas famous brand products.

The emergency power off switch as the standard configuration and comply with the European safety standards.
Electronic load and hydraulic overload protection device.
IP54 Level walking motor is dustproof, waterproof and high in reliability.
The closed electronic control cover plate protects the electrical components free from the eroding of rainwater and dust.
Adopt the AC control system, the motor is maintenance-free, and the anti-sliding function that can't be realized by the series excitation electronic control can be achieved.

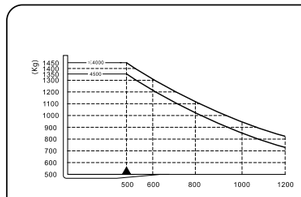
Gantry Specifications

Balanced Weight Electric Forklift
1.5 ton --3.5 ton

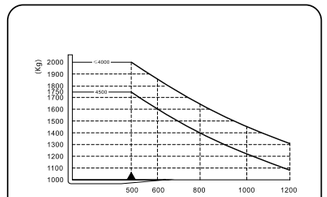
CPD15K-35K Gantry Specifications

Gantry type	Product description	Maximum fork height	Gantry height			Free lifting height		The lowest ground clearance of the gantry	Gantry dip angle		Lifting capacity	
			Landing height of the gantry	Extending height of the gantry		Without load-backrest	With load-backrest		Nose down	Hypsokinesis	500mm load center distance	
				Without load-backrest	With load-backrest						Four supporting points	
		mm	mm	mm	mm	mm	mm	mm	Deg	Deg	Kg	Kg
CPD15K gantry											1.5T	
	Level II standard gantry	3000	2070	3590	3875	35	35	120	6	12	1450	
		3300	2220	3890	4175	35	35	120	6	12	1450	
		3500	2320	4090	4375	35	35	120	6	12	1450	
		3700	2420	4490	4775	35	35	120	6	12	1450	
		4000	2620	4790	5075	35	35	120	6	12	1450	
		4500	2870	5290	5575	35	35	120	6	6	1350	
	Level II free gantry	3000	2040	3590	3875	1475	1005	120	6	12	1500	
		3300	2190	3890	4175	1625	1155	120	6	12	1450	
		3500	2290	4090	4375	1725	1255	120	6	12	1350	
	Level III free gantry	4500	2040	5090	5375	1485	1015	120	6	6	1250	
		4700	2110	5290	5575	1545	1075	120	6	6	1150	
		5000	2190	5590	5875	1685	1215	120	3	3	1000	
		5500	2460	6090	6375	1825	1355	120	3	3	950	
CPD20K/25K gantry											2T	2.5T
	Level II standard gantry	3000	2055	3620	4035	140	140	123	6	12	2000	2500
		3300	2205	3920	4335	140	140	123	6	12	2000	2500
		3500	2305	4120	4535	140	140	123	6	12	2000	2500
		3700	2405	4320	4735	140	140	123	6	12	2000	2500
		4000	2605	4620	5035	140	140	123	6	12	2000	2500
		4500	2855	5120	5535	140	140	123	6	6	1750	2250
	Level II free gantry	3000	2025	3620	4035	1440	1020	123	6	12	2000	2500
		3300	2175	3920	4335	1590	1170	123	6	12	1850	2350
		3500	2275	4120	4535	1690	1270	123	6	12	1800	2250
	Level III free gantry	4500	2055	5120	5535	1470	1050	123	6	6	1550	1950
		5000	2255	5620	6035	1670	1250	123	6	6	1250	1550
		5500	2516	6120	6535	1930	1510	123	3	3	1050	1250
		6000	2708	6620	7035	2123	1703	123	3	3	1100	1100
CPD30K/35K gantry											3T	3.5T
	Level II standard gantry	3000	2145	3680	4285	160	160	145	6	12	3000	3500
		3300	2295	3980	4585	160	160	145	6	12	3000	3500
		3500	2395	4180	4785	160	160	145	6	12	3000	3500
		3700	2495	4380	4985	160	160	145	6	12	3000	3500
		4000	2695	4680	5285	160	160	145	6	12	3000	3500
		4500	2945	5180	5785	160	160	145	6	6	2250	2950
	Level II free gantry	3000	2115	3695	4285	1460	870	145	6	12	3000	3500
		3300	2265	3995	4585	1610	1020	145	6	12	2850	3350
		3500	2365	4195	4785	1710	1120	145	6	12	2750	3250
	Level III free gantry	4500	2115	5195	5785	1460	870	145	6	6	2550	2750
		5000	2315	5695	6285	1660	1070	145	6	6	2150	2450
		5400	2440	6095	6685	1785	1195	145	6	6	1750	1950
		6000	2765	6695	7285	2110	1520	145	3	3	1150	1450

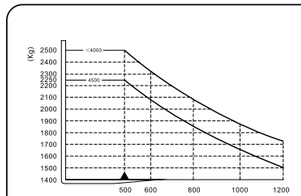
Load Curve Graph



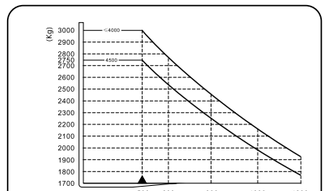
Four supporting points CPD15K load center distance (mm)
Warning: The rated load capacity of the accessories with sidesway needs to decrease 150kg!



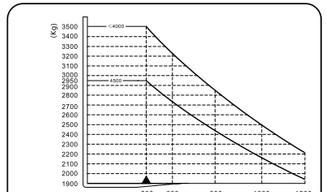
Four supporting points CPD20K load center distance (mm)
Warning: The rated load capacity of the accessories with sidesway needs to decrease 150kg!



Four supporting points CPD25K load center distance (mm)
Warning: The rated load capacity of the accessories with sidesway needs to decrease 200kg!



Four supporting points CPD30K load center distance (mm)
Warning: The rated load capacity of the accessories with sidesway needs to decrease 200kg!



Four supporting points CPD35K load center distance (mm)
Warning: The rated load capacity of the accessories with sidesway needs to decrease 200kg!

Three Color Options



Colors option: the customer can select the color based on the workshop environment

Ice blue:

The main color of Hilectro electric forklift, which stand for the new Haitian generation want to extend the achievements in the new technology field.



Flame orange:

Eye catching color, the best choice for the workshop environment.



Grass green:

The fresh color which means Haitian focus on the green technology.



The colors above should subject to the real painting on the forklift.